

Renesas Device Partition Manager

Renesas Device Partition Manager (RDPM) is a tool embedded into e² studio to assist debugging TrustZone projects by providing TrustZone related features, e.g: DLM state transitioning, secure/non-secure memory boundaries setting, device initialization and current device status retrieval.

Renesas Device Partition Manager

At least one action must be selected

Device Family: Renesas RA

Action

☐ Read current device information
☐ Change debug state
☐ Set TrustZone secure / non-secure boundaries
☐ Initialize device

Target MCU connection: J-Link

Connection Type: SCI

Emulator Connection: Serial No

Serial No/IP Address:

Debugger supply voltage (V): 0

Connection Speed (bps for SCI, Hz for SWD): 9600

Debug state to change to: Secure Software Development

Memory partition sizes

☐ Use Renesas Partition Data file

Browse...

Code Secure (KB)

Code NSC (KB)

Data Start Address (KB)

Data Secure (KB)

SRAM Secure (KB)

SRAM NSC (KB)

Show Command Line
Run

Close

Device Family	Select the device family type.
Action	Select one or more actions to perform.

	Read current device information	Read the current debug state and secure memory boundaries of the device.
	Change debug state	Transition the debug state of the device.
	Set TrustZone secure / non-secure boundaries	Program the secure/non-secure memory boundaries of the device to separate memory into different regions.
	Initialize device	Initialize the device by rolling the debug state back to secure state and reset the memory boundaries to initial values. The flash memory will also be erased.
Target MCU connection	Select the emulator type to connect to the device.	
Connection Type	Select the connection interface to the device. The options are SCI (UART) and SWD.	
Emulator Connection	Select the connection type to the emulator. The options are serial number and IP address. Only Segger J-Link supports this option.	
Serial No/IP Address	Enter the serial number or IP address of the specific emulator that you want to connect your device with.	
Debugger supply voltage (V)	Select the voltage of the power supply to the emulators. Only eligible for E2 and E2 Lite emulators.	
Connection Speed (bps for SCI, Hz for SWD)	Select baud rate (bps) or speed (Hz) of the connection.	
Debug state to change to	Select the target debug state for your device to transition to.	
Memory partition sizes	When "Set TrustZone secure / non-secure boundaries" is selected as one of the actions, specify the size of each memory partition in kilobytes to program the memory boundaries.	
	Use Renesas Partition Data file	Specify a Renesas Partition Data (RPD) file to load the partition sizes.
	Code Secure (KB)	Specify the size of the secure region of code area.
	Code NSC (KB)	Specify the size of the non-secure callable region of code area.
	Data Start Address (KB)	Specify the start address of the data area. Only for supported devices.
	Data Secure (KB)	Specify the size of the secure region of data area.
	SRAM Secure (KB)	Specify the size of the secure region of SRAM.
	SRAM NSC (KB)	Specify the size of the non-secure callable region of SRAM.
Console	Output the result of the actions.	

Show Command Line	Show the actual commands being generated from the RDPM configurations.
Import button	Import RDPM settings.
Export button	Export RDPM settings.

Additional Notes

The purpose of Renesas Device Partition Manager is to provide only the basic and essential TrustZone related features embedded within e² studio to ease the process of TrustZone debugging.

To use other advanced features that are not included in this tool such as those listed below, please download and install the Renesas Flash Programmer (RFP).

- Transition to DLM states other than those available in this tool.
- DLM state or Authentication Level (AL) authentication keys injection.
- Without initializing the device, rollback to previous DLM state or Protection Level (PL) using authentication key method.